

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
GODDARD SPACE FLIGHT CENTER (GSFC)
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
(JOFOC)
FOR PNA-X MICROWAVE NETWORK ANALYZER

1. **Federal Acquisition Regulation (FAR) 6.303-2(b)(1) – Identification of the agency and the contracting activity, and specific identification of the document as a “Justification for other than full and open competition.”**

This document is a justification for other than full and open competition prepared by National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (GSFC).

The procuring agency is the National Aeronautics and Space Administration (NASA) and the contracting activity is NASA’s GSFC, Greenbelt, MD.

2. **FAR 6.303-2(b)(2) – The nature and/or description of the action being approved:**

This justification provides the rationale for contracting by other than full and open competition to award a sole-source contract to Keysight Technologies, Inc. in support of the GSFC Flight Microwave and Telecommunications Branch for a highly specialized engineering test and measurement equipment: a PNA-X microwave network analyzer.

3. **FAR 6.303-2(b)(3) – A description of the supplies or services required, to meet the Agency’s needs (including the estimated value):**

The Keysight PNA-X microwave network analyzer will replace the existing Keysight PNA-X microwave network analyzer in Goddard’s Electromagnetic Anechoic Chamber. This is a highly specialized facility used to measure antenna radiation patterns for flight and ground radio frequency antennas, airborne instrument antennas, research and development antennas, as well as measurements of radio frequency and microwave instrument and devices. This facility has a strict set of functional and performance requirements, as well as custom government owned software that has been developed around the Keysight PNA-X.

The proposed Firm Fixed Price (FFP) contract is anticipated to be awarded in September 2024, with of 6 weeks after award. The estimated value of this procurement is \$358,251.97.

4. **FAR 6.303-2(b)(4) – An identification of the statutory authority permitting other than full and open competition:**

The statutory authority permitting other than full and open competition is 10 U.S.C. 3204(a)(1) as implemented by FAR 6.302-1: Only one responsible source and no other supplies or services will satisfy agency requirements.

5. **FAR 6.303-2(b)(5) – A demonstration that the proposed contractor’s unique qualifications or the nature of the acquisition requires use of the authority cited:**

The rationale supporting the use of 10 U.S.C. 3204(a)(1) is as described below.

The vendor, Keysight Technologies, Inc. has the unique qualifications and specialized skills to provide the network analyzer that meets all of the technical requirements identified in the Statement of Work (SOW).

For readability, the SOW has the technical requirements categorized into software interface requirements, hardware interface requirements, hardware functionality requirements, and hardware performance requirements. The following is a summary rationale for each the category of requirements, and why the Keysight PNA-X described in the SOW is uniquely qualified to meet the requirements.

Software Interfaces: This equipment interfaces with custom government software that controls an antenna test chamber. That custom software has been developed by a team of NASA engineers over two decades and has been written to interface specifically with the Keysight PNA-X series network analyzer. Any other brand or model of network analyzer would require substantial modification of the government software. The technical experts estimate this software modification would cost the government an additional \$250,000.00 and require 12 months of development during which the facility would be out of operation. The Keysight PNA-X is the only model among all competitors that is compatible with the existing government software and will not incur substantial additional cost for software modification.

Hardware Interfaces: This equipment interfaces with existing hardware assets that control an antenna test chamber. Any modification to these hardware interfaces would result in additional cost to the government to modify or procure new hardware for the antenna test chamber. The Keysight PNA-X Model # N5244B is fully compatible to all existing hardware interfaces.

Hardware Functionalities: This equipment operates as the measurement instrumentation for the antenna test chamber, this network analyzer has a series of strict hardware functionality requirements. The Keysight PNA-X Model # N5244B with the options described in the Statement of Work provides all the critical functionalities (i.e. a frequency of operation from 10 MHz to 43.5 GHz, two internal radio sources, and X-parameter measurement capability) and ensures there is no loss in functionality or hardware incompatibility to the antenna test chamber.

Hardware Performance: This equipment operates as the measurement instrumentation for the antenna test chamber, this network analyzer has a series of strict hardware performance

requirements (i.e. Dynamic Range (Extended): ≥ 140 dB up to 35 GHz, ≥ 130 dB above 35 GHz; Maximum Power Output (Leveled): ≥ 15 dBm up to 30 GHz, > 8 dBm above 30 GHz; Power Level Accuracy (Max, both sources): $\leq \pm 2$ dB from 50 MHz to 10, $\leq \pm 2.5$ dB up-to 26.5 GHz, $\leq \pm 3.5$ dB otherwise; Power Level Linearity: $\leq \pm 2$ dB, Power Sweep Range: ≥ 34 dB up-to 30 GHz, ≥ 28 dB otherwise; Input Noise Floor (@ 10 Hz IFBW): ≤ -120 dBm from 500 MHz to 26.5 GHz, ≤ -105 dBm below 500 MHz, ≤ -115 dBm above 26.5 GHz; 0.1 dB Compression Point (Typical): ≥ 15 dBm below 10 GHz, ≥ 13 dBm above 10 GHz). The Keysight PNA-X Model # N5244B with the options described in the Statement of Work meets all performance requirements and ensures there is no loss in the antenna chamber measurement performance.

6. FAR 6.303-2(b)(6) – A description of the efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by Subpart 5.2 and, if not, which exception under 5.202 applies:

A notice to the Government Point of Entry (GPE) website (beta.sam.gov) was published on 07/03/2024, in accordance with FAR Subpart 5.2. This posting informed potential sources of NASA's intent to award this sole-source contract to Keysight Technologies, Inc. The results of this synopsis are summarized in Section 10 below.

7. FAR 6.303-2(b)(7) – A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:

The Contracting Officer's signature on this document indicates that the Contracting Officer has determined that the anticipated cost to the government will be fair and reasonable. The Contractor shall be required to submit a proposal to be evaluated and negotiated by the Government. Prior to execution of the contractual instrument, a proposal analysis will be performed. The proposal analysis will ensure that the final agreed-to price for the contract action is fair and reasonable.

The proposed costs by Keysight will be evaluated by the Contracting Officer's Representative, the Contract Specialist, and the Contracting Officer to determine that the FFP is fair and reasonable. The Contracting Officer will conduct a thorough price analysis of the contractor's proposal inclusive of the proposed terms and conditions.

8. FAR 6.303-2(b)(8) – Description of the market research conducted, and the results, or a statement of the reasons market research was not conducted:

The technical experts in the Flight Microwave and Telecommunication Systems Branch at NASA GSFC conducted market research of network analyzers available in the industry. The commercially available products of five major manufacturers for network analyzers were investigated via published technical reports, catalogs, and equipment data sheets.

The results of the market research revealed the following:

Keysight is the only company that meets the technical requirements of this solicitation to interface to the custom government software.

The summary results of the market research of each of the alternative major manufacturers is described below:

- Siglent Technologies
 - The Siglent Network Analyzer products are not compatible to the software or hardware interfaces of antenna test chamber as described in the statement of work.
- B&K Precision
 - The B&K Precision products are not compatible to the software or hardware interfaces of antenna test chamber as described in the statement of work.
- Anritsu
 - Anritsu offers network analyzer options that meet the hardware interface requirements, but not software interface requirements. Substantial software modification would be required, resulting in additional cost to government and loss of operational readiness for the antenna test chamber while software modification is made.
- Rhode & Schwarz
 - The Rhode & Schwarz high performance network analyzer line meets many but not all the hardware interface requirements. Major substantial hardware re-design and software modification of the antenna test chamber would be required. There would also be a substantial loss in functionality and performance of the antenna test chamber.

Based on this evaluation of current products on the market Keysight is the only responsible source of the network analyzer that will satisfy agency requirements.

9. FAR 6.303-2(b)(9) – Any other facts supporting the use of other than full and open competition:

None.

10. FAR 6.303-2(b)(10) – A listing of the sources, if any, that expressed an interest in writing in the acquisition:

A notice of NASA's intent to award this sole-source action was synopsisized on the GPE website (Sam.gov) per FAR Subpart 5.2 (See Section 6 above). The synopsis was posted on 07/03/2024 and closed on 07/18/2024. On July 8th, two companies made an inquiry to the synopsis. On July 8th, Advance1 Solutions, LLC asked for Network Analyzer part number which was provided. The other inquiry was made by New Tech Solutions Inc to provide a statement of work which was also provided.

11. FAR 6.303-2(b)(11) – A statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:

The Agency will continue to examine the market in the future for alternative solutions or new sources before executing any subsequent acquisitions for the same requirements.

Network Analyzer
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
SIGNATURE PAGE

I certify that the facts presented in this justification are accurate and complete.

A handwritten signature in black ink, appearing to read "Justin Long", is written over a horizontal line.

Justin Long
Technical Representative

I hereby certify that the above justification is complete and accurate to the best of my knowledge and belief.

Teresa S. Shaw
Contracting Officer